
RAG's Hidden Problem

- 1 Vector RAG struggles with precise relationship queries.
- 2 Fails to answer 'global' questions across entities.

The Hybrid RAG Idea

- 1 Combine vectors (semantic search) with graphs (precise relationships).
- 2 LLM acts as a router, directing queries to the right storage.

System Design Flow

1 Ingestion: LLM builds graph, text becomes vectors.

2 Query Routing: Specific facts go to graph, 'vibe' queries to vector DB.

Hybrid Query Router (Python)

- 1 Route based on query type: graph for relationships, vector for semantic.

main.py

```
# Hybrid Query Router (simplified)
def hybrid_rag_query(question: str) -> str:
    if is_relationship_query(question):
        results = graph_db.query(build_cypher(question))
    else:
        results = vector_db.similarity_search(embed(question))
    return llm.generate(f"Combine these facts: {results}")
```

Real-World Impact

- 1 Achieve both high precision (graphs) and high recall (vectors).
- 2 Unlock complex, multi-entity questions that pure vector RAG can't answer.

Final Takeaway

- 1 Don't choose; combine. Hybrid RAG is about precision vs. recall, not purity.
- 2 Follow for more practical AI engineering insights!

